SMIT PATEL

Full Stack & Applied AI Software Engineer | Python, TypeScript, React, Java & AWS

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PROFESSIONAL SUMMARY

Full Stack Software Engineer with **7+ years** of experience designing, building, and scaling enterprise web applications, distributed backend microservices, and automated cloud infrastructure across **IBM**, **Optum**, **Tesla**, and fintech environments. Proven track record of owning features end-to-end across **Python (FastAPI, Flask, Django)**, **TypeScript/React**, and **Java (Spring Boot)** on **AWS** and **Kubernetes**. Experienced in architecting production **AI Agents** and **RAG** workflows—including an autonomous multi-cluster **Incident Triage & Root Cause Analysis (RCA) Agent** deployed across **9 production Kubernetes clusters** at IBM—alongside zero-downtime cloud migrations and GitOps governance.

TECHNICAL SKILLS

- **Languages & Frontend:** Python, TypeScript, JavaScript (ES6+), Java, C#, SQL, Bash, React.js, HTML5, CSS3, Responsive UI/UX Architecture
- **Backend & APIs:** FastAPI, Flask, Django, Spring Boot, Node.js, .NET, RESTful APIs (Synchronous & Asynchronous), Microservices Architecture, GraphQL
- **AI & Agentic Engineering:** Autonomous AI Agents, Tool Calling & Orchestration, Retrieval-Augmented Generation (RAG), LLM API Integration (OpenAI, Gemini, Claude), Automated Incident RCA Workflows, Computer Vision / ML Pipelines
- **Cloud, Kubernetes & DevSecOps:** Amazon Web Services (AWS — EC2, S3, Lambda, RDS, IAM), Microsoft Azure (AZ-900 Certified), Kubernetes (K8s), Docker, Kyverno Policy Governance, VPC Infrastructure, GitOps, Terraform, CI/CD (GitHub Actions, Jenkins), Linux
- **Data, Messaging & Observability:** PostgreSQL, MySQL, SQL Server, Snowflake, Redis, Apache Kafka, OpenSearch, IBM Instana APM, Custom Telemetry & Delay Dashboards

PROFESSIONAL EXPERIENCE

IBM | Lowell, MA — *Nov 2024 – Present*

Software Developer

- **Architected and scaled** an autonomous **AI Incident Triage & Root Cause Analysis (RCA) Agent** across **9 production Kubernetes clusters**, continuously monitoring **Instana** events and **Slack** alert channels to automate first-response diagnostics for SRE and Operations teams.
- **Engineered** multi-source agentic tool-calling and runbook retrieval (**RAG**) workflows that automatically cross-reference internal **wiki runbooks** against live **Instana APM metrics**, custom delay dashboards, **Apache Kafka** consumer lag monitors, **Kubernetes** pod/deployment health, and **OpenSearch** logs to synthesize probable RCAs and drastically reduce manual operational triage toil.
- **Executed** live production cluster migrations from classic infrastructure to **VPC infrastructure**, modernizing the cloud network architecture while maintaining high availability for mission-critical workloads.
- **Implemented** custom **Kyverno** security, compliance, and admission-control policies across **7 production Kubernetes clusters**, enforcing standardized deployment quality gates across engineering squads.
- **Manage** Kubernetes **GitOps** workflows, cloud infrastructure, and production operations within a 5-engineer core platform squad supporting **9 sister engineering teams** under the SIP umbrella.
- **Develop** and maintain backend microservices, asynchronous REST APIs, and automated **CI/CD** pipelines using **Python**, **Node.js**, **Docker**, and **GitHub Actions**.

Optum (UnitedHealth Group) | Boston, MA — *Jan 2023 – Nov 2024*

Software Engineer

- **Delivered** an impactful presentation to a 150-member Data Science Guild, showcasing the utilization of a DCE environment for provisional **Azure Cloud** and **AWS** setups, thereby facilitating R&D in novel data science techniques within Optum.
- **Developed Python**-based **microservices** to orchestrate **Snowflake** resources for internal tenants, facilitating **CRUD operations** on the platform.
- **Successfully elevated** a fellow Trainee Development Program (TDP) associate from a non-technical background to **Java** proficiency through dedicated mentorship, covering comprehensive OOP concepts and industry best practices, while simultaneously enhancing our team's project contributions.
- **Spearheaded** the development of resilient **Java Spring Boot** APIs and pioneered the integration of automated code quality checks using **GitHub Actions** within the PAAN application, substantially improving code reliability and deployment efficiency by **20%**.
- **Innovated** within the HCP platform by designing and executing PRM agent resources and **Infrastructure as Code**, significantly advancing operational effectiveness, and by architecting a new **Snowflake** tenancy model using **Python** that bolstered the adoption of **CI/CD** and DevOps practices, setting a new standard for project scalability and collaboration.

Tesla, Inc. | Remote / Palo Alto, CA *Apr 2022 – Jan 2023*
Software Engineer (Python)

- **Revolutionized** the deployment process by constructing and deploying **CI/CD** pipelines with **GitHub Actions**, **Jenkins**, **Docker**, and **Kubernetes**, cutting deployment times by half and boosting overall productivity through streamlined and automated workflows.
- **Led** a critical system upgrade from **Python 2.7 to Python 3**, expertly transitioning databases from **PostgreSQL** to **MySQL** with strategic **Linux** SED scripting, thereby maintaining system integrity and enhancing security, and further secured application access by implementing **OAuth 2.0** and **OIDC** with **Python 3** and **Django** for reliable **JWT** handling and protected user authentication flows.
- **Initiated** the initiative to upgrade Tesla's Open edX **Django** application to **Python 3**, fortifying the educational platform's security and performance, and integrated dynamic content management with **Node.js** and **Akamai Storage** solutions, enabling efficient handling of over **50** remote learning modules.

Lendica, Inc. | Boston, MA *Feb 2021 – Oct 2021*
Software Engineer

- **Upgraded** Lendica's Application Portal by engineering an advanced automation system with **Python** and **OAuth 2.0**, drastically improving the efficiency of third-party application integrations and streamlining the user experience.
- **Designed** and deployed a highly responsive data processing pipeline using **Apache Airflow 2.0**, enabling automated, condition-based triggering of **REST API** functions that accelerated data retrieval and updates within the **AWS S3** environment.
- **Played** a crucial role in forging a productive partnership with Terrayn, a top-tier Digital Marketing & SEO firm, by co-developing an MIT-licensed **Python pip package** that supports Metrc compliance, complemented by an **npm** frontend package, significantly enhancing industry-standard regulatory adherence.
- **Showcased** extensive **AWS** proficiency by employing **boto** and **boto3** libraries to facilitate efficient cloud operations and crafted a secure **OAuth 2.0** authentication protocol for Lendica's front-facing platforms, incorporating automated backend token generation and **PostgreSQL** encryption to uphold high-security standards.

Acespritech Pvt. Ltd. | Gandhinagar, Gujarat, IN *Oct 2017 – Sep 2019*
Assoc. Software Developer

- **Excelled** in the creation of application components utilizing advanced programming techniques within an **Agile** framework, resulting in the timely delivery of high-quality software solutions.
- **Developed** responsive and user-centric web interfaces with **Python** and **Django**, improving user experience and contributing to a **20%** increase in development efficiency through proactive involvement in code testing.

SELECTED AI & FULL STACK PROJECTS

Autonomous Multi-Cluster AI SRE & Incident RCA Agent | *Python, LLMs, RAG, Kubernetes, Kafka, OpenSearch, Instana*

- Built an event-driven production **AI agent** integrated with **Slack** alerts and observability APIs across **9 Kubernetes clusters** to automate runbook lookup, metric/log correlation, **Kafka** lag inspection, and probable RCA generation for Operations engineers.

Adaptive Routine & Dynamic Scheduling Full Stack Web Application | *Python (REST APIs), React, TypeScript, LLM APIs*

- Engineered an end-to-end full stack web application featuring dynamic scheduling algorithms, state-driven **React/TypeScript** UI components, relational persistence, and **LLM API** workflows for real-time schedule optimization from user activity logs.

Real-Time Facial Expression Recognition & Computer Vision API | *Python, Machine Learning, Computer Vision, Flask*

- Implemented a real-time facial emotion detection and classification ML pipeline in **Python**, exposing low-latency model inference endpoints via a **Flask REST API** integrated with an interactive web interface.

EDUCATION

University of Massachusetts Boston | Boston, MA — **Master of Science (M.S.) in Computer Science** *Graduated Dec 2021 | GPA: 3.8*

LDRP-ITR | Gandhinagar, GUJ, IN — **Bachelor of Engineering (B.E.) in Computer Engineering** *Graduated 2017 | GPA: 3.5*